WORKED SOLUTION

STEP I 2005, Question 3

Pure • Solutions of equation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q3

The key idea

Multiply through by $(x-a)(x-b)$ to turn each equation into a polynomial equation, then count its real roots. The point that needs care is that clearing denominators could, in principle, add "solutions" $x=a$ or $x=b$ that the original equation doesn't allow, so we must check that this never happens. In part (ii) the quadratic has a repeated root exactly when its discriminant is zero, and that gives the condition.

Solution

Part (i)

For $x \neq a, b$ we may multiply by $(x-a)(x-b)$, and the equation is equivalent to

$$\begin{aligned} x(x-b)+x(x-a) &= (x-a)(x-b) \\ 2x^2-(a+b)x &= x^2-(a+b)x+ab \\ x^2 &= ab. \end{aligned}$$

Since a and b have the same sign, $ab>0$, so this has the two distinct real roots $x=\sqrt{ab}$ and $x=-\sqrt{ab}$.

We still need these to be allowed values. If $x=a$ were a root, then $a^2=ab$, and since $a\neq 0$ this gives $a=b$, which is false. In the same way $x=b$ is not a root. So both roots are genuine solutions, and the equation has exactly two distinct real solutions, $x=\pm\sqrt{ab}$.

Part (ii)

Again, for $x \neq a, b$ multiply by $(x-a)(x-b)$:

$$2x^2-(a+b)x = (1+c)(x^2-(a+b)x+ab),$$

which rearranges to

$$q(x) \equiv (1-c)x^2 + c(a+b)x - (1+c)ab = 0.$$

Because $c \neq 1$, the leading coefficient is non-zero, so this really is a quadratic.

Ask yourself Could one of the roots of q be a or b, so that it has to be thrown away?

Substitute and see:

$$q(a) = (1-c)a^2 + c(a+b)a - (1+c)ab = a^2 - ab = a(a-b),$$
$$q(b) = b^2 - ab = b(b-a)$$

(the c terms cancel). Both are non-zero because a, b are non-zero and distinct. So a and b are never roots of q. This means every real root of q is a solution of the original equation, and the reverse is also true.

The discriminant of q is

$$\begin{aligned}\Delta &= c^2(a+b)^2 + 4(1-c)(1+c)ab \\ &= c^2[(a+b)^2 - 4ab] + 4ab \\ &= c^2(a-b)^2 + 4ab.\end{aligned}$$

If $c^2 = -\dfrac{4ab}{(a-b)^2}$ (and $a \neq b$, so we may divide by $(a-b)^2$), then $\Delta = 0$. The quadratic then has exactly one (repeated) real root,

$$x = -\frac{c(a+b)}{2(1-c)},$$

and by the check above this root is not a or b. So the equation has exactly one real solution, as required. (In fact the converse holds as well. If $\Delta > 0$ there are two solutions and if $\Delta < 0$ there are none, so the condition is also necessary.)

Rewriting the condition. Over a common denominator,

$$1 - \left(\frac{a+b}{a-b}\right)^2 = \frac{(a-b)^2 - (a+b)^2}{(a-b)^2} = \frac{-4ab}{(a-b)^2},$$

so the condition is $c^2 = 1 - \left(\dfrac{a+b}{a-b}\right)^2$, as required.

Deducing $0 < c^2 \leq 1$. The square $\left(\frac{a+b}{a-b}\right)^2$ is non-negative, so the second form gives $c^2 \leq 1$. The first form gives $c^2 = -4ab/(a-b)^2$. This is non-zero because $ab \neq 0$, and $c^2 \geq 0$ for real c, so $c^2 > 0$. (This also tells us that $ab < 0$: a and b have opposite signs, which fits with part (i).) Hence the condition can only hold if $0 < c^2 \leq 1$.

The upper value is attained. $c^2 = 1$ exactly when $a + b = 0$, and then $c = -1$ (since $c \neq 1$).

Checks

- Take $a = 4$, $b = -1$. Then $c^2 = 16/25$; choose $c = \frac{4}{5}$. The repeated root is $x = -\frac{4}{5} \cdot 3/(2 \cdot \frac{1}{5}) = -6$, and indeed $\frac{-6}{-10} + \frac{-6}{-5} = 0.6 + 1.2 = 1.8 = 1 + c$.
- The extreme case $a = 1$, $b = -1$, $c = -1$ gives $\frac{x}{x-1} + \frac{x}{x+1} = 0$, i.e. $2x^2 = 0$: one solution, $x = 0$.
- In part (i), $a = 1$, $b = 4$ gives $x = \pm 2$: $\frac{2}{1} + \frac{2}{-2} = 1$ and $\frac{-2}{-3} + \frac{-2}{-6} = 1$.

Common slips

- Forgetting to check that the roots are not a or b after clearing denominators. Here it turns out to be harmless, but it has to be shown.
- Expanding the discriminant without grouping the c^2 terms. Writing $(a+b)^2 - 4ab = (a-b)^2$ is what makes the condition fall out neatly.
- Claiming $c^2 \geq 0$ instead of $c^2 > 0$. The strict inequality comes from $ab \neq 0$.

Beyond STEP

The left-hand side is a rational function, and "how many times does its graph meet the horizontal line $y = 1 + c$?" is a question about its range. The condition found here picks out the turning-point values of that function, the heights where a horizontal line just touches the graph. Asking when a polynomial has a repeated root leads to the discriminant and, more generally, to the resultant, which appear in first-year algebra. For more problems in this style, see Siklos, *Advanced Problems in Mathematics*.

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